```
import numpy as np
#create arrays
a = np.array([1, 2, 3, 4])
b = np.zeros((2,3))
c = np.ones((2,3))
d = np.random.rand(2,3)
print(a)
print(b)
print(c)
print(d)
```

```
[1 2 3 4]
[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]]
[[0.75870779 0.56048547 0.73439118]
 [0.61635861 0.71970461 0.0989605 ]]
```

```
# array operations
print(a.shape)
print(a.mean)
print(a.std())
print(np.sum(a))
print(np.max(a))
```

```
(4,)
<built-in method mean of numpy.ndarray object at 0x7eb4913d5c50>
1.118033988749895
10
4
```

```
# matrix opertions
x = np.array([[1, 2], [3, 4]])
y = np.array([[5, 6], [7, 8]])
print(np.dot(x, y))
print(np.transpose(x))
```

```
[[19 22]
 [43 50]]
[[1 3]
 [2 4]]
```

```
import pandas as pd
# create DataFrame
data = {
    'Age': [25, 30, 45],
    'Salary': [30000, 40000, 50000],
    'Department':['IT', 'HR', 'Finance']
}
df = pd.DataFrame(data)
print(df.describe())
```

```
             Age   Salary
count   3.000000      3.0
mean   33.333333  40000.0
std    10.408330  10000.0
min    25.000000  30000.0
25%    27.500000  35000.0
50%    30.000000  40000.0
75%    37.500000  45000.0
max    45.000000  50000.0
```

Gemini

```
# Handling missing values
df['Salary'].fillna(df['Salary'].mean(), inplace=True)
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())
```

```
/tmp/ipykernel_674/2408465813.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through ch
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['Salary'].fillna(df['Salary'].mean(), inplace=True)
```

```
import pandas as pd
# encoding categorial data
data = {
    'Age': [25, 30, 45]
```

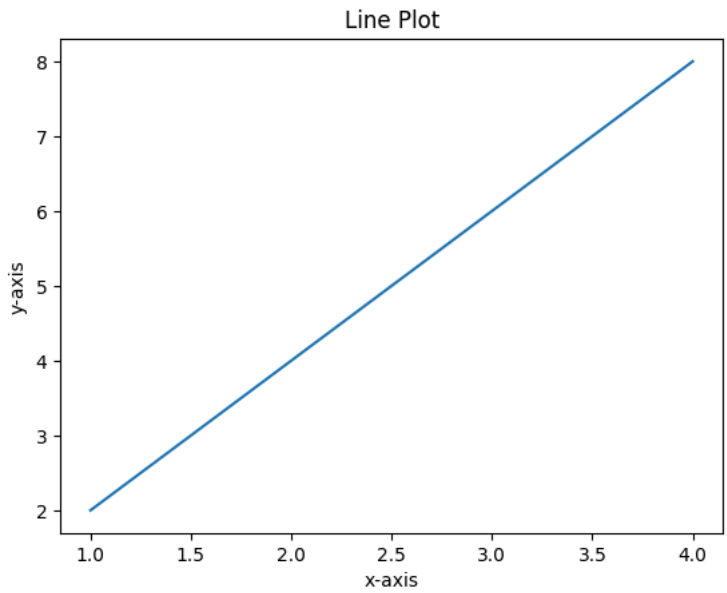
```
    'Salary': [30000, 40000, 50000],
    'Department':['IT', 'HR', 'Finance']
}
df_encoded = pd.get_dummies(df, drop_first=True)
print(df_encoded)
```

```
   Age  Salary  Department_HR  Department_IT
0   25   30000          False           True
1   30   40000           True          False
2   45   50000          False          False
```

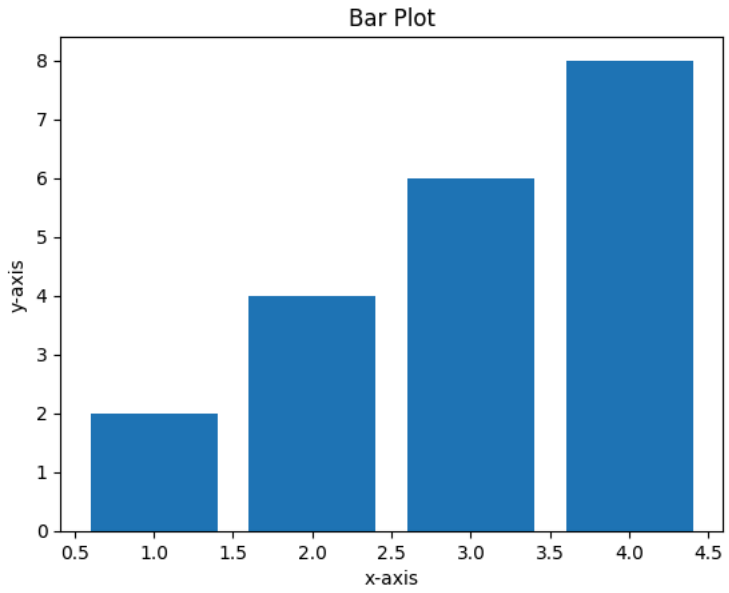
```
df_encoded = pd.get_dummies(df, drop_first=True)
print(df_encoded)
```

```
   Age  Salary  Department_HR  Department_IT
0   25   30000          False           True
1   30   40000           True          False
2   45   50000          False          False
```

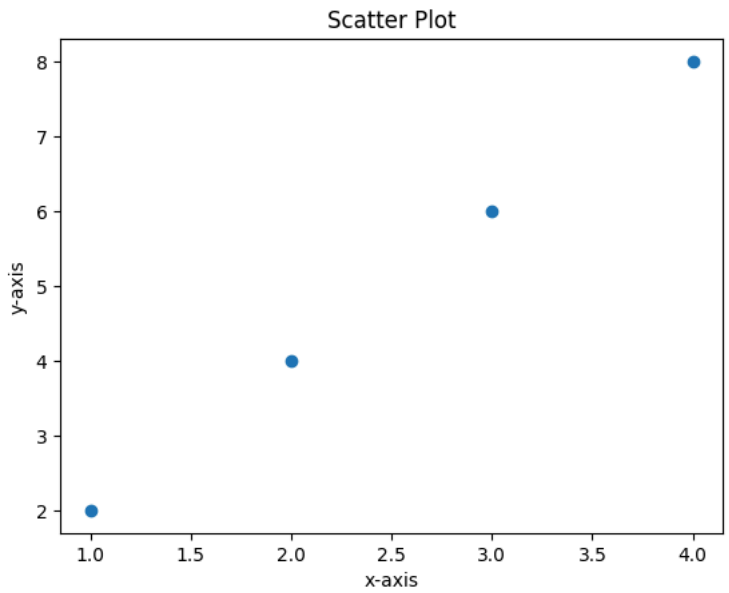
```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4]
y = [2, 4, 6, 8]
plt.plot(x, y)
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Line Plot')
plt.show()
```



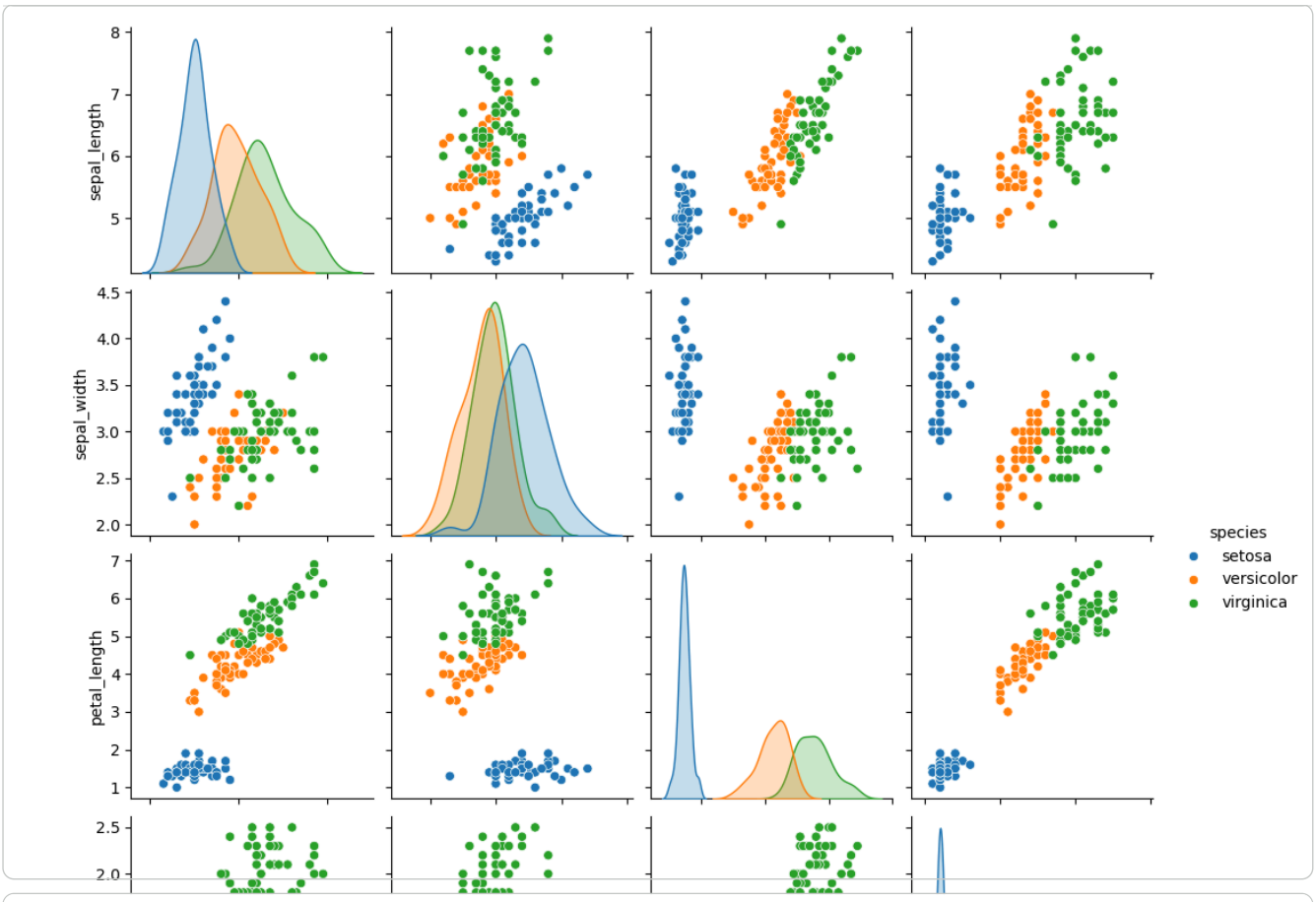

```
# bar plot
plt.bar(x, y)
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Bar Plot')
plt.show()
```



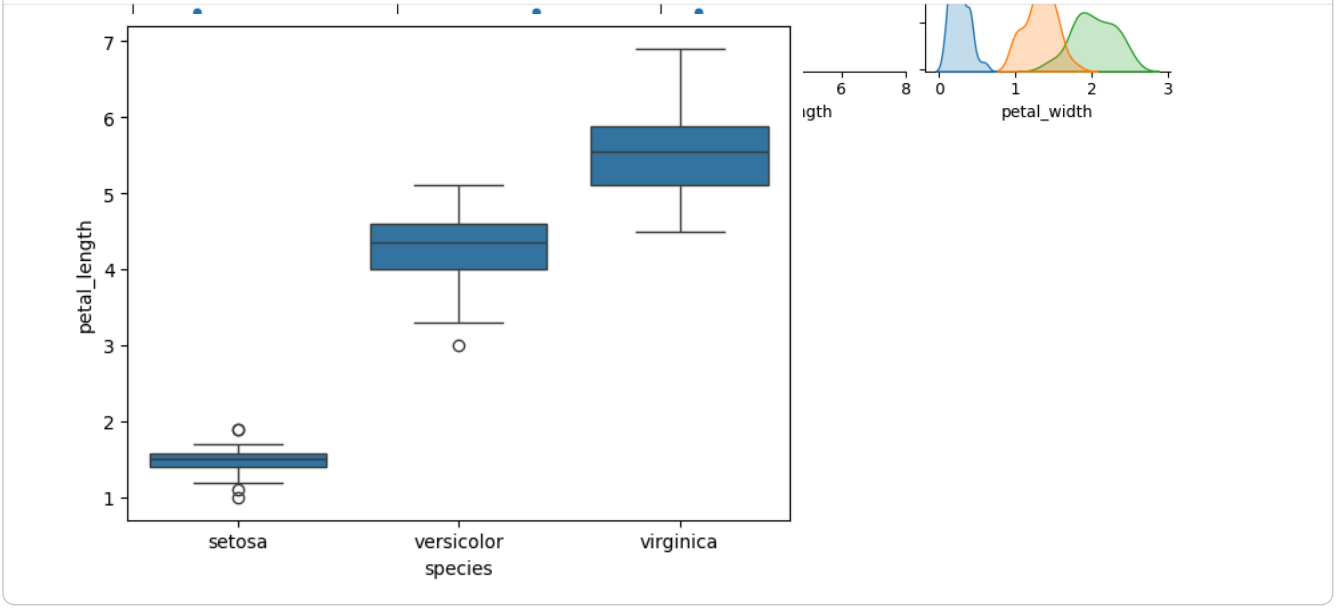

```
# scatter plot
plt.scatter(x, y)
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Scatter Plot')
plt.show()
```




```
import seaborn as sns
import matplotlib.pyplot as plt
# load sample dataset
df = sns.load_dataset('iris')
# create pair plot
sns.pairplot(df, hue='species')
plt.show()
```




```
# Box plot
sns.boxplot(x='species', y='petal_length', data=df)
plt.show()
```




```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```